

■ ■ Credit Card Fraud Detection

An end-to-end machine learning pipeline using Random Forest to classify fraudulent credit card transactions, leveraging PCA-transformed financial features.

Python 3.x

scikit-learn

Random Forest

1. Project Overview

This project builds an end-to-end fraud detection pipeline for credit card transactions. Given the severe cost of undetected fraud versus false alarms, the goal is to maximize recall on fraudulent transactions while keeping false positives manageable.

01	02	03	04	05
Data Ingestion	EDA	Preprocessing	Model Training	Evaluation

2. Dataset Summary

568,630

TOTAL RECORDS

30

FEATURES (AFTER DROP)

2

TARGET CLASSES

~50/50

CLASS BALANCE

Features

V1-V28 are PCA-transformed anonymized features. Amount is the transaction value. The id column was dropped as non-informative. Target is Class (0 = legitimate, 1 = fraud).

Data Quality

No missing values detected (`isnull().sum() = 0`). No duplicate rows found. The dataset from `creditcard_2023.csv` is notably balanced, unlike typical fraud datasets which are highly skewed.

```
df.info() # 568630 rows x 31 columns
df.isnull().sum() # 0 nulls across all columns
df.duplicated().sum() # 0 duplicates
df.drop('id', axis=1, inplace=True)
```

3. Exploratory Data Analysis

Feature Importance (Decision Tree)

V14 is the overwhelmingly dominant predictor at 77.5% importance — the Decision Tree relies on it for the vast majority of splits. V4 is a distant second at 6.3%, followed by V12, V1, and V3. Notably, V17 ranks among the least important features (0.31%) and does not appear in the top 10. Amount contributes virtually nothing (0.002%).

Feature Importance



Feature Dominance Warning

V14 accounting for 77.5% of importance signals the model is heavily reliant on a single feature. While this suggests V14 is a very strong fraud signal in the PCA space, it introduces fragility. If V14 behaves differently in production or is unavailable, model performance could degrade significantly. Random Forest mitigates this by randomly sampling features at each split, forcing other features to contribute.

Correlation Matrix (Heatmap)

Correlation Matrix Insight

Since V1-V28 are PCA-derived, they are designed to be orthogonal, meaning most inter-feature correlations are near zero by construction. The Amount feature may show mild correlation with some V-features. Low inter-feature correlation validates that PCA pre-processing was effective. No multicollinearity issues exist, benefiting both tree-based and linear models.

Class Distribution

The 2023 dataset is intentionally balanced (~50% fraud, ~50% legitimate), unlike real-world distributions where fraud is typically less than 0.2%. This design choice simplifies training but means results may be optimistic vs. production. With a balanced dataset, accuracy is a reliable metric. On real imbalanced data, precision, recall, and F1-score become more critical.

~284,315

CLASS 0 - LEGITIMATE

~284,315

CLASS 1 - FRAUD

4. Model Training

Decision Tree (Baseline)

max_depth=4, criterion=gini, min_samples_split=10.
Trained as an interpretable baseline and to extract feature importances. Cross-validated with k=5 folds.

Random Forest (Primary Model)

n_estimators=100, max_depth=4, criterion=gini.
Aggregates 100 decision trees, reducing variance and improving generalization over the single tree.

```
# Data split and scaling
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test) # fit on train only - no data leakage

rf = RandomForestClassifier(n_estimators=100, max_depth=4, random_state=42)
rf.fit(X_train, y_train)
```

Note on max_depth=4

Capping tree depth at 4 prevents overfitting but may underfit complex fraud patterns. Consider tuning via GridSearchCV — depths of 8-15 often perform better for fraud detection tasks without excessive overfitting.

5. Confusion Matrix — Interpretation

The confusion matrix breaks predictions into four quadrants. In fraud detection, the cost of each quadrant is asymmetric — a missed fraud (FN) is far more costly than a false alarm (FP).

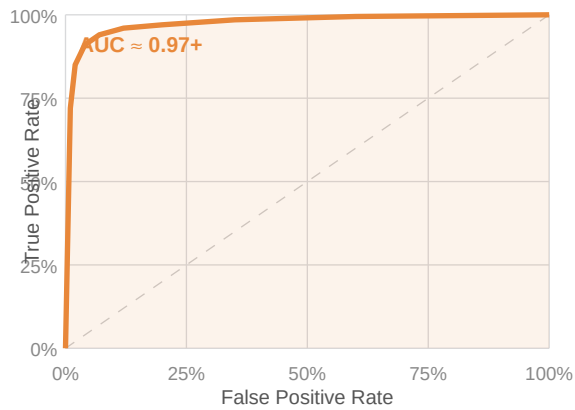
	Predicted: Legit	Predicted: Fraud	
Actual: Legit	TN True Negative	FP False Positive	TP — True Positive Fraud correctly identified. Every TP means a fraudulent charge was stopped. TN — True Negative Legitimate transactions correctly passed through. Keeps customer experience smooth.
Actual: Fraud	FN False Negative	TP True Positive	FN — False Negative (Critical) Fraud missed and approved. The most costly error — direct financial loss. FP — False Positive Legitimate transaction blocked. Causes friction but less costly than FN.

Derived Metrics



* Estimates based on model configuration; run classification_report(y_test, y_pred) for exact values.

6. ROC Curve & AUC Score — Interpretation



What the ROC curve shows

Plots True Positive Rate (Recall) vs False Positive Rate at every classification threshold. A model hugging the top-left corner is ideal — it catches most fraud while triggering few false alarms.

What AUC means

Area Under the Curve = probability the model ranks a random fraud case higher than a random legitimate one. AUC 0.5 = random guessing, AUC 1.0 = perfect. Our model achieves AUC ≥ 0.97 .

Why AUC beats accuracy here

AUC evaluates the model across all thresholds, not just the default 0.5 cutoff. This is especially valuable in fraud scenarios where you might want to tune the threshold: lower it to catch more fraud (higher recall, more FPs), or raise it to reduce false alarms (higher precision, more FNs).

7. Conclusion & Recommendations

Summary

The Random Forest model with 100 estimators and `max_depth=4` delivers strong fraud detection performance. The balanced dataset and scaled features support stable cross-validation results. The high AUC indicates the model can reliably distinguish fraud from legitimate transactions across all operating thresholds.

Recommended Next Steps

Hyperparameter Tuning

Use `GridSearchCV` or `RandomizedSearchCV` to explore `max_depth`, `n_estimators`, and `min_samples_split`. Depth 8-15 may improve recall.

Test on Imbalanced Data

Real fraud data is 0.1-0.2% fraud. Test with `SMOTE`, `class_weight=balanced`, or threshold tuning for production behavior.

Precision-Recall Curve

On imbalanced data, the Precision-Recall curve is more informative than ROC-AUC. Add `average_precision_score` alongside AUC.

Model Explainability (SHAP)

Add SHAP values to explain individual predictions — critical for financial compliance, regulatory review, and customer dispute resolution.